Bianchi identities

Special Mathematics Lectures – Differential Geometry

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Definition 1. For a smooth manifold M and an affine connection ∇, the torsion and the curvature of the connection are respectively defined by

$$T(X,Y) := \nabla_X Y - \nabla_Y X - [X,Y] \ \in \mathfrak{X}(M), \tag{1}$$

and

$$R(X,Y) := \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X,Y]} \ \in \mathrm{End}(\mathfrak{X}(M)), \tag{2}$$

for any $X, Y \in \mathfrak{X}(M)$.

The maps $(X,Y) \mapsto T(X,Y)$ and $(X,Y,Z) \mapsto R(X,Y)Z$ are $C^\infty(M)$-linear in their arguments.

Proposition 2. *Let X, Y, Z be smooth vector fields on a smooth manifold M with an affine connection ∇.*

(i) The torsion $T(X,Y)$ is $C^\infty(M)$-linear in X and Y.

(ii) The curvature $R(X,Y)Z$ is $C^\infty(M)$-linear in X, Y, Z.

Proof. (i) For any $f, g, h \in C^\infty(M)$ and $X, Y \in \mathfrak{X}(M)$, we have

$$\begin{aligned}
[fX, gY](h) =& fX(gY(h)) - gY(fX(h)) \\
=& fX(g)Y(h) + gfX(Y(h)) - gY(f)X(h) - fgY(X(h)) \\
=& fg\left(X(Y(h)) - Y(X(h))\right) + fX(g)Y(h) - gY(f)X(h) \\
=& fg[X,Y](h) + fX(g)Y(h) - gY(f)X(h).
\end{aligned} \tag{3}$$

Therefore, we have

$$[fX, gY] = fg[X,Y] + fX(g)Y - gY(f)X \tag{4}$$

$$\Leftrightarrow \quad [fX, gY] - fX(g)Y + gY(f)X = fg[X,Y]. \tag{5}$$

Using the definition of the torsion, we get

$$\begin{aligned}
T(fX, gY) &= \nabla_{fX} gY - \nabla_{gY} fX - [fX, gY] \\
&= f\nabla_X(gY) - g\nabla_Y(fX) - [fX, gY] \\
&= fX(g)Y + fg\nabla_X Y - gY(f)X - gf\nabla_Y X - [fX, gY] \\
&= fg\nabla_X Y - gf\nabla_Y X - ([fX, gY] - fX(g)Y + gY(f)X) \\
&= fg\nabla_X Y - gf\nabla_Y X - fg[X,Y] \\
&= fgT(X,Y).
\end{aligned} \tag{6}$$

In addition, it is to check that $T(X, Y+Z) = T(X,Y) + T(X,Z)$ and $T(X+Y, Z) = T(X,Z) + T(Y,Z)$ for any $X, Y, Z \in \mathfrak{X}(M)$. Consequently, $T(X,Y)$ is $C^\infty(M)$-linear in X and Y.

(ii) For any $f, g, h \in C^\infty(M)$ and $X, Y, Z \in \mathfrak{X}(M)$, we have

$$\begin{aligned}
R(fX, gY)hZ &= \nabla_{fX}\nabla_{gY} hZ - \nabla_{gY}\nabla_{fX} hZ - \nabla_{[fX,gY]} hZ \\
&= f\nabla_X(g\nabla_Y hZ) - g\nabla_Y(f\nabla_X hZ) - \nabla_{[fX,gY]} hZ.
\end{aligned} \tag{7}$$

For the first two terms, we have

$$
\begin{aligned}
& f\nabla_X(g\nabla_Y hZ) - g\nabla_Y(f\nabla_X hZ) \\
&= fX(g)\nabla_Y hZ + fg\nabla_X\nabla_Y hZ - gY(f)\nabla_X hZ - fg\nabla_Y\nabla_X hZ \\
&= \nabla_{fX(g)Y - gY(f)X} hZ + fg\nabla_X\nabla_Y hZ - fg\nabla_Y\nabla_X hZ \\
&= \nabla_{[fX,gY]-fg[X,Y]} hZ + fg\nabla_X\nabla_Y hZ - fg\nabla_Y\nabla_X hZ \\
&= \nabla_{[fX,gY]} hZ + fg(\nabla_X\nabla_Y hZ - \nabla_Y\nabla_X hZ - \nabla_{[X,Y]} hZ).
\end{aligned} \tag{8}
$$

Therefore, we get

$$
R(fX, gY)hZ = fg(\nabla_X\nabla_Y hZ - \nabla_Y\nabla_X hZ - \nabla_{[X,Y]} hZ). \tag{9}
$$

Now, we have

$$
\begin{aligned}
& \nabla_X\nabla_Y hZ - \nabla_Y\nabla_X hZ - \nabla_{[X,Y]} hZ \\
&= \nabla_X(Y(h)Z + h\nabla_Y Z) - \nabla_Y(X(h)Z + h\nabla_X Z) - [X,Y](h)Z - h\nabla_{[X,Y]} Z \\
&= X(Y(h))Z + Y(h)\nabla_X Z + X(h)\nabla_Y Z + h\nabla_X\nabla_Y Z \\
&\quad - Y(X(h))Z - X(h)\nabla_Y Z - Y(h)\nabla_X Z - h\nabla_Y\nabla_X Z \\
&\quad - [X,Y](h)Z - h\nabla_{[X,Y]} Z \\
&= \left(X(Y(h)) - Y(X(h)) - [X,Y](h)\right) Z + h(\nabla_X\nabla_Y Z - h\nabla_Y\nabla_X Z - \nabla_{[X,Y]} Z) \\
&= hR(X,Y)Z.
\end{aligned} \tag{10}
$$

As a result, we get

$$
R(fX, gY)hZ = fghR(X,Y)Z. \tag{11}
$$

In addition, it is easy to check that

$$
R(X, Y+Z)W = R(X,Y)W + R(X,Z)W, \tag{12}
$$

$$
R(X+Y, Z)W = R(X,Z)W + R(Y,Z)W, \tag{13}
$$

$$
R(X,Y)(Z+W) = R(X,Y)Z + R(X,Y)W, \tag{14}
$$

for any $X, Y, Z, W \in \mathfrak{X}(M)$. □

Due to the $C^\infty(M)$-linearity of the torsion and the curvature, the torsion and the curvature are tensor fields on the manifold M. Since $T(X,Y)$ is a vector field, which is also a tensor field in $\mathcal{T}_1{}^0(M)$, the torsion T is a tensor field in $\mathcal{T}_1{}^2(M)$. Let (U, φ) be a chart and $\{E_j\}_{j=1}^m$ be the corresponding vector fields defined by $(E_j)_p = E_{j,p}$ for $p \in U$. Setting $T(E_i, E_j)^k = T_{i,j}{}^k$, we have

$$
T(X,Y) = \sum_{i=1}^m \sum_{j=1}^m \sum_{k=1}^m X^i Y^j T_{i,j}{}^k E_k. \tag{15}
$$

Since $[E_i, E_j] = 0$, we get

$$
T_{i,j}{}^k = \mathrm{d}x^k\left(T(E_i, E_j)\right) = \mathrm{d}x^k\left(\nabla_{E_i} E_j\right) - \mathrm{d}x^k\left(\nabla_{E_j} E_i\right) = \Gamma^k_{i,j} - \Gamma^k_{j,i}. \tag{16}
$$

Similarly, since $R(X,Y)Z$ is a vector field, which is also a tensor field in $\mathcal{T}_1{}^0(M)$, the curvature R is a tensor field in $\mathcal{T}_1{}^3(M)$. Setting $\left(R(E_i, E_j)E_k\right)^\ell = R_{i,j,k}{}^\ell$, we have

$$
R(X,Y)Z = \sum_{i=1}^m \sum_{j=1}^m \sum_{k=1}^m \sum_{\ell=1}^m X^i Y^j Z^k R_{i,j,k}{}^\ell E_\ell. \tag{17}
$$

Using $[E_i, E_j] = 0$, we get

$$\begin{aligned}
R_{i,j,k}{}^{\ell} &= \mathrm{d}x^{\ell}\left(R(E_i, E_j)E_k\right) = \mathrm{d}x^{\ell}\left(\nabla_{E_i}\nabla_{E_j}E_k\right) - \mathrm{d}x^{\ell}\left(\nabla_{E_j}\nabla_{E_i}E_k\right) \\
&= \mathrm{d}x^{\ell}\left(\nabla_{E_i}\sum_{r=1}^{m}\Gamma^{r}_{j,k}E_r\right) - \mathrm{d}x^{\ell}\left(\nabla_{E_j}\sum_{r=1}^{m}\Gamma^{r}_{i,k}E_r\right) \\
&= \sum_{r=1}^{m}E_i(\Gamma^{r}_{j,k})\,\mathrm{d}x^{\ell}\left(E_r\right) + \sum_{r=1}^{m}\Gamma^{r}_{j,k}\,\mathrm{d}x^{\ell}\left(\nabla_{E_i}E_r\right) \\
&\quad - \sum_{r=1}^{m}E_j(\Gamma^{r}_{i,k})\,\mathrm{d}x^{\ell}\left(E_r\right) - \sum_{r=1}^{m}\Gamma^{r}_{i,k}\,\mathrm{d}x^{\ell}\left(\nabla_j(E_r)\right) \\
&= E_i(\Gamma^{\ell}_{j,k}) + \sum_{r=1}^{m}\Gamma^{r}_{j,k}\Gamma^{\ell}_{i,r} - E_j(\Gamma^{\ell}_{i,k}) - \sum_{r=1}^{m}\Gamma^{r}_{i,k}\Gamma^{\ell}_{j,r} \\
&= \frac{\partial\Gamma^{\ell}_{j,k}}{\partial x^i} - \frac{\partial\Gamma^{\ell}_{i,k}}{\partial x^j} + \sum_{r=1}^{m}\left(\Gamma^{r}_{j,k}\Gamma^{\ell}_{i,r} - \Gamma^{r}_{i,k}\Gamma^{\ell}_{j,r}\right).
\end{aligned} \tag{18}$$

Remark. For a Riemannian manifold, the connection is called metric-compatible if the following identity holds for any vector fields $X, Y, Z \in \mathfrak{X}(M)$:

$$X(\phi(Y,Z)) = \phi(\nabla_X Y, Z) + \phi(Y, \nabla_X Z), \tag{19}$$

where ϕ is the metric tensor field of the Riemannian manifold.

If the torsion of the connection is zero, then the connection is called torsion-free and the curvature satisfies the following identity.

Lemma 3 (The first Bianchi identity)**.** *Assume that the connection on a smooth manifold is torsion free. Then, the following identity holds for the curvature: for any $X, Y, Z \in \mathfrak{X}(M)$,*

$$R(X,Y)Z + R(Y,Z)X + R(Z,X)Y = 0, \tag{20}$$

where 0 means the 0-vector field

Proof. The condition of having a torsion-free connection on the manifold implies that for any vector fields $X, Y \in \mathfrak{X}(M)$, we have

$$T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y] = 0. \tag{21}$$

Differentiating this identity with respect to a vector field $Z \in \mathfrak{X}(M)$ gives

$$\nabla_Z\nabla_X Y - \nabla_Z\nabla_Y X - \nabla_Z[X,Y] = 0. \tag{22}$$

Now, using the torsion-free condition again, we have

$$T([X,Y],Z) = \nabla_{[X,Y]}Z - \nabla_Z[X,Y] - [[X,Y],Z] = 0. \tag{23}$$

Or equivalently,

$$\nabla_Z[X,Y] = \nabla_{[X,Y]}Z - [[X,Y],Z]. \tag{24}$$

Substitute (24) into (22), we get

$$\nabla_Z\nabla_X Y - \nabla_Z\nabla_Y X - \nabla_{[X,Y]}Z + [[X,Y],Z] = 0. \tag{25}$$

Interchanging $X \leftrightarrow Y$, $Y \leftrightarrow Z$, $Z \leftrightarrow X$ and repeting the same process, we get

$$\nabla_X\nabla_Y Z - \nabla_X\nabla_Z Y - \nabla_{[Y,Z]}X + [[Y,Z],X] = 0. \tag{26}$$

Repeting the same process, we also get

$$\nabla_Y\nabla_Z X - \nabla_Y\nabla_X Z - \nabla_{[Z,X]}Y + [[Z,X],Y] = 0. \tag{27}$$

Adding (25), (26), (27), and reorganizing terms gives

$$\begin{aligned}
&\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z \\
&+ \nabla_Y \nabla_Z X - \nabla_Z \nabla_Y X - \nabla_{[Y,Z]} X \\
&+ \nabla_Z \nabla_X Y - \nabla_X \nabla_Y Y - \nabla_{[Z,X]} Y \\
&+ [[X,Y],Z] + [[Y,Z],X] + [[Z,X],Y] = 0.
\end{aligned} \tag{28}$$

We simplify this equation by using the definition of the curvature

$$R(X,Y)Z + R(Y,Z)X + R(Z,X)Y + [[X,Y],Z] + [[Y,Z],X] + [[Z,X],Y] = 0. \tag{29}$$

Furthermore, the Jacobi identity gives

$$[[X,Y],Z] + [[Y,Z],X] + [[Z,X],Y] = 0. \tag{30}$$

Consequently, we obtain the first Bianchi identity. □

As a result, the curvature of a Riemannian connection (torsion-free and metric-compatible) always satisfies the first Bianchi identity. If $\phi \in \Sigma^2(M)$ is the metric tensor field of the Riemannian manifold M, then the curvature tensor field $R \in \mathcal{T}^4(M)$ is defined by

$$R(X,Y,Z,W) = \phi(R(X,Y)Z, W), \tag{31}$$

for $X, Y, Z, W \in \mathfrak{X}(M)$. The curvature tensor field R satisfies the following properties.

Lemma 4. *For any Riemannian manifold (M, ϕ) with a Riemannian connection, the following equalities hold:*

(i) $R(X,Y,Z,W) + R(Y,Z,X,W) + R(Z,X,Y,W) = 0$,

(ii) $R(X,Y,Z,W) = -R(Y,X,Z,W)$,

(iii) $R(X,Y,Z,W) = -R(X,Y,W,Z)$,

(iv) $R(X,Y,Z,W) = R(Z,W,X,Y)$.

Proof. (i) By Lemma 3, we have

$$\phi\big(R(X,Y)Z + R(Y,Z)X + R(Z,X)Y,\ W\big) = \phi(0, W) = 0. \tag{32}$$

By the linearity of the metric tensor, we get

$$\phi\left(R(X,Y)Z, W\right) + \phi\left(R(Y,Z)X, W\right) + \phi\left(R(Z,X)Y, W\right) = 0. \tag{33}$$

Or equivalently,

$$R(X,Y,Z,W) + R(Y,Z,X,W) + R(Z,X,Y,W) = 0. \tag{34}$$

(ii) Since $\nabla_{[X,Y]} Z = \nabla_{-[Y,X]} Z = -\nabla_{[Y,X]} Z$, we have from the definition of the curvature,

$$\begin{aligned}
R(X,Y)Z &= \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z \\
&= -\left(\nabla_Y \nabla_X Z - \nabla_X \nabla_Y Z - \nabla_{[Y,X]} Z\right) \\
&= -R(Y,X)Z.
\end{aligned} \tag{35}$$

Therefore, we get

$$R(X,Y,Z,W) = \phi(R(X,Y)Z, W) = -\phi(R(Y,X)Z, W) = -R(Y,X,Z,W). \tag{36}$$

(iii) First of all, we have

$$\begin{aligned}
R(X,Y,Z,W) &= \phi(R(X,Y)Z, W) \\
&= \phi(\nabla_X \nabla_Y Z, W) - \phi(\nabla_Y \nabla_X Z, W) - \phi(\nabla_{[X,Y]} Z, W).
\end{aligned} \tag{37}$$

Using the metric-compatibility of the connection and the symmetric property of the metric tensor, we have

$$\begin{aligned}\phi(\nabla_X\nabla_Y Z, W) &= X(\phi(\nabla_Y Z, W)) - \phi(\nabla_Y Z, \nabla_X W)\\ &= X(\phi(\nabla_Y Z, W)) - Y(\phi(Z, \nabla_X W)) + \phi(Z, \nabla_Y\nabla_X W)\\ &= X(\phi(\nabla_Y Z, W)) - Y(\phi(Z, \nabla_X W)) + \phi(\nabla_Y\nabla_X W, Z).\end{aligned} \tag{38}$$

Similarly, we also have

$$\phi(\nabla_Y\nabla_X Z, W) = Y(\phi(\nabla_X Z, W)) - X(\phi(Z, \nabla_Y W)) + \phi(\nabla_X\nabla_Y W, Z). \tag{39}$$

Therefore, we get

$$\begin{aligned}&\phi(\nabla_X\nabla_Y Z, W) - \phi(\nabla_Y\nabla_X Z, W)\\ &= X\big(\phi(\nabla_Y Z, W) + \phi(Z, \nabla_Y W)\big) - Y\big(\phi(\nabla_X Z, W) + \phi(\nabla_X Z, W)\big)\\ &\quad + \phi\big(\nabla_Y\nabla_X W - \nabla_X\nabla_Y W,\ Z\big)\\ &= X\left(Y(\phi(Z,W))\right) - Y\left(X(\phi(Z,W))\right) - \phi\big(\nabla_X\nabla_Y W - \nabla_Y\nabla_X W,\ Z\big)\\ &= [X,Y](\phi(Z,W)) - \phi\big(\nabla_X\nabla_Y W - \nabla_Y\nabla_X W,\ Z\big)\\ &= \phi(\nabla_{[X,Y]}Z, W) + \phi(Z, \nabla_{[X,Y]}W) - \phi\big(\nabla_X\nabla_Y W - \nabla_Y\nabla_X W,\ Z\big)\\ &= \phi(\nabla_{[X,Y]}Z, W) - \phi\big(\nabla_X\nabla_Y W - \nabla_Y\nabla_X W - \nabla_{[X,Y]}W,\ Z\big)\\ &= \phi(\nabla_{[X,Y]}Z, W) - \phi(R(X,Y)W, Z).\end{aligned} \tag{40}$$

As a result, we get

$$R(X,Y,Z,W) = -\phi(R(X,Y)W, Z) = -R(X,Y,W,Z). \tag{41}$$

(iv) Using (i) four times, we have

$$\begin{aligned}R(X,Y,Z,W) + R(Y,Z,X,W) + R(Z,X,Y,W) &= 0,\\ R(Y,Z,W,X) + R(Z,W,Y,X) + R(W,Y,Z,X) &= 0,\\ R(Z,W,X,Y) + R(W,X,Z,Y) + R(X,Z,W,Y) &= 0,\\ R(W,X,Y,Z) + R(X,Y,W,Z) + R(Y,W,X,Z) &= 0.\end{aligned} \tag{42}$$

Then, using (iii), we get

$$\begin{aligned}R(X,Y,Z,W) - R(Y,Z,W,X) + R(Z,X,Y,W) &= 0,\\ R(Y,Z,W,X) - R(Z,W,X,Y) + R(W,Y,Z,X) &= 0,\\ R(Z,W,X,Y) - R(W,X,Y,Z) - R(X,Z,Y,W) &= 0,\\ R(W,X,Y,Z) - R(X,Y,Z,W) - R(Y,W,Z,X) &= 0.\end{aligned} \tag{43}$$

Adding all these equations gives

$$R(Z,X,Y,W) - R(X,Z,Y,W) + R(W,Y,Z,X) - R(Y,W,Z,X) = 0. \tag{44}$$

Using (ii), we get

$$2R(Z,X,Y,W) - 2R(Y,W,Z,X) = 0 \quad \Leftrightarrow \quad R(Z,X,Y,W) = R(Y,W,Z,X). \qquad \square$$

All properties of R in Lemma 4 can be written in terms of the components $R_{i,j,k,\ell} = R(E_i, E_j, E_k, E_\ell)$ of the curvature tensor field R:

(i) $R_{i,j,k,\ell} + R_{j,k,i,\ell} + R_{k,i,j,\ell} = 0$,

(ii) $R_{i,j,k,\ell} = -R_{j,i,k,\ell}$,

(iii) $R_{i,j,k,\ell} = -R_{i,j,\ell,k}$,

(iv) $R_{i,j,k,\ell} = R_{k,\ell,i,j}$,

for every $i, j, k, \ell \in \{1, \dots, m\}$. These properties restrict the number of independent components $R_{i,j,k,\ell}$ of the curvature tensor field. The anti-symmetric property (ii) implies that there are only $n := \binom{m}{2}$ independent pairs of indices (i, j). Similarly, the anti-symmetric property (iii) also implies that there are only n independent pairs of indices (k, ℓ). Next, the symmetric property (iv) implies that there are only $n + \binom{n}{2}$ or $n(n+1)/2$ independent pairs (i, j) and (k, ℓ). Furthermore, all these properties (i)-(iv) imply that the totally anti-symmetric part of curvature tensor R vanishes, i.e. $\mathscr{A}R = 0$. For $m < 4$, there always exists a pair of indices with the same value, so this constraint is not restrictive. Hence, the number of independent components of the curvature tensor field R for $m < 4$ is F(m)=n(n+1)/2, where $n = \binom{m}{2}$. For $m \geq 4$, the constraint $\mathscr{A}R = 0$ gives $\binom{m}{4}$ independent equations corresponding to all possible 4 distinct indices i, j, k, ℓ. Therefore, the number of independent components of the curvature tensor field R for $m \geq 4$ is

$$\begin{aligned} F(m) &= \frac{n(n+1)}{2} - \binom{m}{4} \\ &= \frac{1}{2}\frac{m(m-1)}{2}\left(\frac{m(m-1)}{2} + 1\right) - \frac{1}{24}m(m-1)(m-2)(m-3) \\ &= \frac{1}{8}(m^4 - 2m^3 + 3m^2 - 2m) - \frac{1}{24}(m^4 - 6m^3 + 11m^2 - 6m) \\ &= \frac{1}{12}m^2(m^2 - 1). \end{aligned} \tag{45}$$

For $m = 1$, since $n = \binom{1}{2} = 0$, the number of independent components of the curvature tensor field R is 0. This means that a one-dimensional Riemannian manifold with a Riemannian connection has no *intrinsic* curvature. Without a larger embedding space, every one-dimensional Riemannian manifold is flat. Otherwise, it is possible to define a non-zero extrinsic curvature (e.g. curvature in the Frenet frame) for curves embedded in a higher dimensional manifold. For $m = 2$, since $n = \binom{2}{2} = 1$, the number of independent components of the curvature tensor field R is 1. This means that the curvature of a two-dimensional Riemannian manifold with a Riemannian connection is completely determined by a single component. Famously, in general relativity which describes a four-dimensional pseudo-Riemannian manifold where the positive-definite condition of ϕ is dropped, the number of independent components of the curvature tensor field R is $F(4) = 20$.

The following identity is important in the formulation of the Einstein field equations in general relativity.

Proposition 5 (The second Bianchi identity)**.** *Assume that the connection on a smooth manifold is torsion free. For any vector fields $X, Y, Z \in \mathfrak{X}(M)$, the following identity holds*

$$(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0. \tag{46}$$

Proof. Let $X, Y, Z, W \in \mathfrak{X}(M)$. Using the fact that the connection is torsion-free, i.e. $T(X, Y) = 0$, we have

$$\begin{aligned} 0 = R\big(T(X, Y), Z\big)W &= R\big(\nabla_X Y - \nabla_Y X - [X, Y],\ Z\big)W, \\ &= R(\nabla_X Y, Z)W - R(\nabla_Y X, Z)W - R([X, Y], Z)W. \end{aligned} \tag{47}$$

Similarly, we have

$$\begin{aligned} R(\nabla_Y Z, X)W - R(\nabla_Z Y, X)W - R([Y, Z], X)W = 0, \\ R(\nabla_Z X, Y)W - R(\nabla_X Z, Y)W - R([Z, X], Y)W = 0. \end{aligned} \tag{48}$$

Taking the sum of these equations gives

$$\begin{aligned} &R(\nabla_X Y, Z)W - R(\nabla_Y X, Z)W - R([X, Y], Z)W \\ &+ R(\nabla_Y Z, X)W - R(\nabla_Z Y, X)W - R([Y, Z], X)W \\ &+ R(\nabla_Z X, Y)W - R(\nabla_X Z, Y)W - R([Z, X], Y)W = 0. \end{aligned} \tag{49}$$

Using the property $R(X, Y)Z = -R(Y, X)Z$, we get

$$\begin{aligned} &R(\nabla_X Y, Z)W + R(Y, \nabla_X Z)W - R([Y, Z], X)W \\ &+ R(\nabla_Y Z, X)W + R(Z, \nabla_Y X)W - R([Z, X], Y)W \\ &+ R(\nabla_Z X, Y)W + R(X, \nabla_Z Y)W - R([X, Y], Z)W = 0. \end{aligned} \tag{50}$$

Using the Leibniz rule (see Appendix), we have

$$\nabla_X(R(Y,Z)W) = (\nabla_X R)(Y,Z)W + R(\nabla_X Y,Z)W + R(Y,\nabla_X Z)W + R(Y,Z)\nabla_X W, \quad (51)$$
$$\nabla_Y(R(Z,X)W) = (\nabla_Y R)(Z,X)W + R(\nabla_Y Z,X)W + R(Z,\nabla_Y X)W + R(Z,X)\nabla_Y W, \quad (52)$$
$$\nabla_Z(R(X,Y)W) = (\nabla_Z R)(X,Y)W + R(\nabla_Z X,Y)W + R(X,\nabla_Z Y)W + R(X,Y)\nabla_Z W. \quad (53)$$

Using these equations, we get

$$\begin{aligned}
&\nabla_X(R(Y,Z)W) - (\nabla_X R)(Y,Z)W - R(Y,Z)\nabla_X W - R([Y,Z],X)W \\
&+ \nabla_Y(R(Z,X)W) - (\nabla_Y R)(Z,X)W - R(Z,X)\nabla_Y W - R([Z,X],Y)W \\
&+ \nabla_Z(R(X,Y)W) - (\nabla_Z R)(X,Y)W - R(X,Y)\nabla_Z W - R([X,Y],Z)W = 0.
\end{aligned} \quad (54)$$

Now, using the definition of the curvature tensor field R, we get

$$\begin{aligned}
&\nabla_Z(R(X,Y)W) - R(X,Y)\nabla_Z W - R([X,Y],Z)W \\
&= \nabla_Z\left(\nabla_X\nabla_Y W - \nabla_Y\nabla_X W - \nabla_{[X,Y]}W\right) \\
&\quad - \left(\nabla_X\nabla_Y\nabla_Z W - \nabla_Y\nabla_X\nabla_Z W - \nabla_{[X,Y]}\nabla_Z W\right) \\
&\quad - \left(\nabla_{[X,Y]}\nabla_Z W - \nabla_Z\nabla_{[X,Y]}W - \nabla_{[[X,Y],Z]}W\right) \\
&= \nabla_Z(\nabla_X\nabla_Y - \nabla_Y\nabla_X)W - (\nabla_X\nabla_Y - \nabla_Y\nabla_X)\nabla_Z W + \nabla_{[[X,Y],Z]}W \\
&= \nabla_Z[\nabla_X,\nabla_Y]W - [\nabla_X,\nabla_Y]\nabla_Z W + \nabla_{[[X,Y],Z]}W \\
&= [\nabla_Z,[\nabla_X,\nabla_Y]]W + \nabla_{[[X,Y],Z]}W.
\end{aligned} \quad (55)$$

Mapulating the other terms in the same way and substituting them into the equation (54), we get

$$\begin{aligned}
&(\nabla_X R)(Y,Z)W + (\nabla_Y R)(Z,X)W + (\nabla_Z R)(X,Y)W \\
&= \big([\nabla_Z,[\nabla_X,\nabla_Y]] + [\nabla_X,[\nabla_Y,\nabla_Z]] + [\nabla_Y,[\nabla_Z,\nabla_X]]\big)W + \nabla_{[[X,Y],Z]+[[Y,Z],X]+[[Z,X],Y]}W \\
&= 0,
\end{aligned} \quad (56)$$

where we have used the Jacobi identity in the last step, i.e.

$$[[X,Y],Z] + [[Y,Z],X] + [[Z,X],Y] = 0, \quad (57)$$
$$[\nabla_Z,[\nabla_X,\nabla_Y]] + [\nabla_X,[\nabla_Y,\nabla_Z]] + [\nabla_Y,[\nabla_Z,\nabla_X]] = 0. \quad (58)$$

The RHS of the first equation is the 0-vector field, while the RHS of the second equation is a map from any vector field to the 0-vector field. □

For $X = E_i$, $Y = E_j$, $Z = E_k$, and $W = E_\ell$, the second Bianchi identity can be written in terms of the components of the curvature R as follows:

$$(\nabla_i R)_{j,k,\ell}{}^n + (\nabla_j R)_{k,i,\ell}{}^n + (\nabla_k R)_{i,j,\ell}{}^n = 0. \quad (59)$$

Note that the covariant derivative preserves the symmetries of the curvature R.

Appendix

Covariant derivative of tensor fields

A smooth function f can be considered as a $(0,0)$-tensor field in the set $\mathcal{T}_0{}^0(M) := C^\infty(M)$ on a smooth manifold M. It is natural to define the covariant derivative of a function f with respect to a vector field $X \in \mathfrak{X}(M)$ by

$$\nabla_X f := X(f). \tag{60}$$

Next, the definition of the covariant derivative of a $(1,0)$-tensor field $\omega \in \mathcal{T}_0{}^1(M)$ with respect to a vector field $X \in \mathfrak{X}(M)$ is given as follows.
Definition 6. Let M be a smooth manifold and ∇ be an affine connection on M. The covariant derivative $\nabla_X \omega \in \mathcal{T}_0{}^1(M)$ of a 1-form field $\omega \in \mathcal{T}_0{}^1(M)$ with respect to a vector field $X \in \mathfrak{X}(M)$ is defined by

$$(\nabla_X \omega)(Y) := X(\omega(Y)) - \omega(\nabla_X Y), \tag{61}$$

for any $Y \in \mathfrak{X}(M)$.

This definition is given so that the Leibniz rule holds. Indeed, Let (U, φ) be a chart and $\{E_j\}_{j=1}^m$ be the corresponding vector fields defined by $(E_j)_p = E_{j,p}$ for $p \in U$. Then, we can define a set of 1-form fields $\{\mathrm{d}x^j\}_{j=1}^m$ where $\{(\mathrm{d}x^j)_p\}_{j=1}^m$ is the dual basis at each point $p \in U$. For any 1-form field $\omega \in \mathcal{T}_0{}^1(M)$ and any vector field $X \in \mathfrak{X}(M)$, we have

$$\omega(X) = \sum_{j=1}^m \omega_j \,\mathrm{d}x^j \left(\sum_{k=1}^m X^k E_k \right) = \sum_{j=1}^m \sum_{k=1}^m \omega_j X^k \,\mathrm{d}x^j\,(E_k) = \sum_{j=1}^m \sum_{k=1}^m \omega_j X^k \delta_{jk} = \sum_{j=1}^m \omega_j X^j. \tag{62}$$

Now, expressing the definition 6 in terms of components, we have

$$\sum_{i=1}^m (\nabla_X \omega)_i Y^i = \sum_{i=1}^m X(\omega_i Y^i) - \omega_i (\nabla_X Y)^i = \sum_{i=1}^m \nabla_X (\omega_i Y^i) - \omega_i (\nabla_X Y)^i \tag{63}$$

$$\Leftrightarrow \quad \sum_{i=1}^m \nabla_X(\omega_i Y^i) = \sum_{i=1}^m (\nabla_X \omega)_i Y^i + \omega_i (\nabla_X Y)^i. \tag{64}$$

Hence, the covariant derivative of a product follows the Leibniz rule. Now, define $\nabla_{E_i} := \nabla_i$. Then, we have for $X = E_i$ and $Y = E_j$,

$$(\nabla_{E_i} \omega)_j = E_i(\omega_j) - \sum_{k=1}^m \omega_k (\nabla_{E_i} E_j)^k. \tag{65}$$

Or, equivalently,

$$(\nabla_i \omega)_j = E_i(\omega_j) - \sum_{k=1}^m \Gamma_{ij}^k \omega_k = \frac{\partial \omega_j}{\partial x^i} - \sum_{k=1}^m \Gamma_{ij}^k \omega_k, \tag{66}$$

where $\left(\frac{\partial \omega_j}{\partial x^i}\right)_p$ means that $\frac{\partial}{\partial x^i}\left(\omega_j \circ \varphi^{-1}\right)(\varphi(p))$. For a dual basis $\omega = \mathrm{d}x^j$, we get

$$\nabla_i \,\mathrm{d}x^j = \sum_{k=1}^m (\nabla_i \,\mathrm{d}x^j)_k \,\mathrm{d}x^k = - \sum_{k=1}^m \Gamma_{ik}^j \,\mathrm{d}x^k\,. \tag{67}$$

Next, the definition of the covariant derivative can be generalized to the case of an arbitrary tensor field $\phi \in \mathcal{T}_s{}^r(M)$.

Definition 7. Let M be a smooth manifold and ∇ be an affine connection on M. The covariant derivative $\nabla_X \phi \in \mathcal{T}_s{}^r(M)$ of a tensor field $\phi \in \mathcal{T}_s{}^r(M)$ with respect to a vector field $X \in \mathfrak{X}(M)$ is defined by

$$\begin{aligned}
&(\nabla_X \phi)(Y_1, \dots, Y_r, \omega_1, \dots, \omega_s) \\
&= X(\phi(Y_1, \dots, Y_r, \omega_1, \dots, \omega_s)) - \sum_{i=1}^{r} \phi(Y_1, \dots, \nabla_X Y_i, \dots, Y_r, \omega_1, \dots, \omega_s) \\
&\quad - \sum_{i=1}^{s} \phi(Y_1, \dots, Y_r, \omega_1, \dots, \nabla_X \omega_i, \dots, \omega_s),
\end{aligned} \tag{68}$$

for any $Y_1, \dots, Y_r \in \mathfrak{X}(M)$ and $\omega_1, \dots, \omega_s \in \mathcal{T}_0{}^1(M)$.

For $X = E_i$, $Y_\ell = E_{j_\ell}$, and $\omega_q = \mathrm{d}x^{k_q}$, we have

$$\begin{aligned}
&(\nabla_{E_i} \phi)(E_{j_1}, \dots, E_{j_r}, \mathrm{d}x^{k_1} \dots \mathrm{d}x^{k_s}) \\
&= E_i(\phi(E_{j_1}, \dots, E_{j_r}, \mathrm{d}x^{k_1} \dots \mathrm{d}x^{k_s})) - \sum_{\ell=1}^{r} \phi(E_{j_1}, \dots, \nabla_{E_i} E_{j_\ell}, \dots, E_{j_r}, \mathrm{d}x^{k_1}, \dots, \mathrm{d}x^{k_s}) \\
&\quad - \sum_{q=1}^{s} \phi(E_{j_1}, \dots, E_{j_r}, \mathrm{d}x^{k_1}, \dots, \nabla_{E_i} \mathrm{d}x^{k_q}, \dots, \mathrm{d}x^{k_s}).
\end{aligned} \tag{69}$$

Using $\nabla_{E_i} E_{j_\ell} = \sum_{p=1}^{m} \Gamma^p_{i,j_\ell} E_p$, and $\nabla_{E_i} \mathrm{d}x^{k_q} = -\sum_{p=1}^{m} \Gamma^{k_q}_{i,p} \mathrm{d}x^p$,

and $\phi(E_{j_1}, \dots, E_{j_r}, \mathrm{d}x^{k_1} \dots \mathrm{d}x^{k_s}) = \phi_{j_1,\dots,j_r}{}^{k_1,\dots,k_s}$, we get

$$\begin{aligned}
(\nabla_i \phi)_{j_1,\dots,j_r}{}^{k_1,\dots,k_s} &= E_i(\phi_{j_1,\dots,j_r}{}^{k_1,\dots,k_s}) - \sum_{\ell=1}^{r} \sum_{p=1}^{m} \Gamma^p_{i,j_\ell} \phi_{j_1,\dots,j_{\ell-1},p,j_{\ell+1},\dots,j_r}{}^{k_1,\dots,k_s} \\
&\quad + \sum_{q=1}^{s} \sum_{p=1}^{m} \Gamma^{k_q}_{i,p} \phi_{j_1,\dots,j_r}{}^{k_1,\dots,k_{q-1},p,k_{q+1},\dots,k_s}.
\end{aligned} \tag{70}$$

Remark. The metric-compatible condition of the connection implies that the metric tensor field $\phi \in \Sigma^2(M)$ satisfies

$$\nabla_X \phi(Y, Z) = X(\phi(Y, Z)) - \phi(\nabla_X Y, Z) - \phi(Y, \nabla_X Z) = 0, \tag{71}$$

for any $X, Y, Z \in \mathfrak{X}(M)$. Or equivalently, $\nabla_X \phi = 0$ for any $X \in \mathfrak{X}(M)$.

From these definitions, we see that the covariant derivative in general is defined as a multilinear map $\nabla : \mathfrak{X}(M) \times \mathcal{T}_s{}^r(M) \to \mathcal{T}_s{}^r(M)$ satisfying for any $X \in \mathfrak{X}(M)$, $\phi \in \mathcal{T}_s{}^r(M)$, and $f \in C^\infty(M)$,

(i) $\nabla_{fX} \phi = f \nabla_X \phi$,

(ii) $\nabla_X (f\phi) = X(f)\phi + f \nabla_X \phi$.

We can easily check these properties using the definitions given above. In the proof of the second Bianchi identity, we have used the Leibniz rule for the covariant derivative of the curvature. To clarify that calculation, we first note that a vector field is also a $(0,1)$-tensor field, which belongs to $\mathcal{T}_1{}^0(M)$, in the sense that for any $X \in \mathfrak{X}(M)$ and any $\omega \in \mathcal{T}_0{}^1(M)$, we have

$$X(\omega) := \omega(X). \tag{72}$$

Now, for a moment, denote $R(X, Y)Z$ by $\mathrm{R}(X, Y, Z, \,\cdot\,)$, which is an element of $\mathcal{T}_1{}^0(M)$. Then, we have for $X, Y, Z, W \in \mathfrak{X}(M)$ and $\omega \in \mathcal{T}_0{}^1(M)$,

$$\begin{aligned}
(\nabla_X \mathrm{R})(Y, Z, W, \omega) &= X\big(\mathrm{R}(Y, Z, W, \omega)\big) - \mathrm{R}(\nabla_X Y, Z, W, \omega) - \mathrm{R}(Y, \nabla_X Z, W, \omega) \\
&\quad - \mathrm{R}(Y, Z, \nabla_X W, \omega) - \mathrm{R}(Y, Z, W, \nabla_X \omega).
\end{aligned} \tag{73}$$

In addition, since $\mathrm{R}(X,Y,Z,\ \cdot\ ) \in \mathcal{T}_1{}^0(M)$, we also have

$$\big(\nabla_X \mathrm{R}(Y,Z,W,\ \cdot\ )\big)(\omega) = X\big(\mathrm{R}(Y,Z,W,\omega)\big) - \mathrm{R}(Y,Z,W,\nabla_X\omega). \tag{74}$$

Therefore, we get

$$(\nabla_X \mathrm{R})(Y,Z,W,\omega) = \big(\nabla_X \mathrm{R}(Y,Z,W,\ \cdot\ )\big)(\omega) - \mathrm{R}(\nabla_X Y,Z,W,\omega) - \mathrm{R}(Y,\nabla_X Z,W,\omega) \tag{75}$$
$$- \mathrm{R}(Y,Z,\nabla_X W,\omega). \tag{76}$$

Since this is true for any $\omega \in \mathcal{T}_0{}^1(M)$, we have

$$(\nabla_X \mathrm{R})(Y,Z,W,\ \cdot\ ) = \nabla_X \mathrm{R}(Y,Z,W,\ \cdot\ ) - \mathrm{R}(\nabla_X Y,Z,W,\ \cdot\ ) - \mathrm{R}(Y,\nabla_X Z,W,\ \cdot\ ) \tag{77}$$
$$- \mathrm{R}(Y,Z,\nabla_X W,\ \cdot\ ). \tag{78}$$

Changing the notation back, we get

$$(\nabla_X R)(Y,Z)W = \nabla_X\big(R(Y,Z)W\big) - R(\nabla_X Y,Z)W - R(Y,\nabla_X Z)W - R(Y,Z)\nabla_X W. \tag{79}$$

Or equivalently,

$$\nabla_X\big(R(Y,Z)W\big) = (\nabla_X R)(Y,Z)W + R(\nabla_X Y,Z)W + R(Y,\nabla_X Z)W + R(Y,Z)\nabla_X W. \tag{80}$$

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